Problem 4

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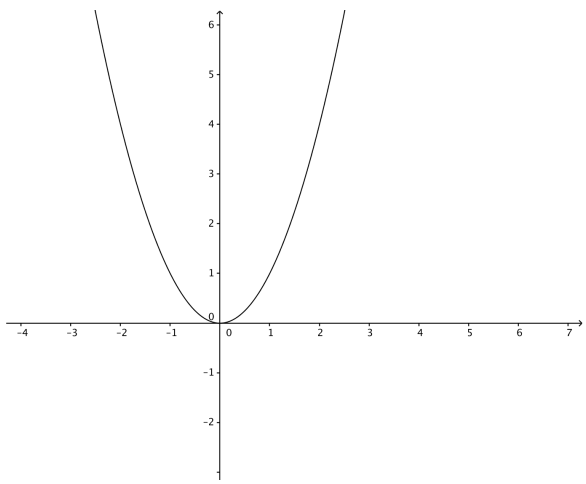
Problem 1 *The graph of a function f is the parabola given in the figure below.*



(a) *Find a formula for $f(x)$.*

(b) *Suppose F_1 is an anti-derivatives of f satisfying $F_1(0) = 2$. Sketch and label the graph of F_1.*
Sketch the graphs of three more antiderivatives of f.

(c) *Using the expression you found in part (a) and your sketches in part (b), find the algebraic representation of F_1.*

(d) *Suppose we're given $h(x) = (1/3)x^3 + 17$, what is $h'(x)$?*

(e) *What is the relationship between f, F_1, h, and h'?*